

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I SWETA R (192511408) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

ANSWERS:

Mock Interview – Big Data

1. Explain the difference between HDFS and traditional file systems.

Answer: HDFS is designed for **distributed storage of very large datasets** across multiple machines. It divides files into blocks and stores them across DataNodes with replication. Traditional file systems usually store data on a single machine or centralized storage system and are designed for smaller-scale workloads. HDFS provides **fault tolerance, scalability, and high-throughput access**.

2. Why is replication important in distributed storage systems?

Answer: Replication creates multiple copies of data on different nodes. If one node fails, another copy can be accessed, preventing data loss and improving availability. In HDFS, replication also helps maintain reliable storage even when hardware failures occur.

3. How does MapReduce divide work between map and reduce phases?

Answer: The **Map phase** processes input data and produces intermediate **key-value pairs**. The framework then performs **shuffle and sort** to group values with the same key. The **Reduce phase** processes these grouped values and generates the final output. This division allows processing to happen in parallel across multiple nodes.

4. What role does YARN play in large-scale Hadoop processing?

Answer: YARN acts as Hadoop's **resource management and job scheduling layer**. It manages CPU and memory resources across the cluster and allocates them to applications. This allows multiple processing frameworks and jobs to share the same Hadoop cluster efficiently.

5. Differentiate NAS and SAN in an enterprise data environment.

Answer:

NAS	SAN
Provides file-level storage	Provides block-level storage
Accessed over a network using file protocols	Usually uses dedicated storage networks
Easier to manage and share files	Better suited for high-performance applications
Common for file sharing and backups	Common for databases and enterprise applications

6. What is the purpose of ETL in a big data pipeline?

Answer: ETL stands for **Extract, Transform, Load**. It extracts data from different sources, transforms it by cleaning and standardizing it, and loads it into a target system such as a data warehouse or data lake. ETL improves **data quality and consistency** before analysis.

7. How does Apache NiFi help in data integration workflows?

Answer: Apache NiFi provides a **visual, flow-based platform** for moving and processing data between different systems. It supports data routing, transformation, filtering, monitoring, and error handling. Its visual interface makes complex data flows easier to design and monitor.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer: Common failure points include **DataNode failures, NameNode failures, network failures, and disk failures**. I would handle them using HDFS replication, NameNode high availability, monitoring, automatic recovery, and regular backups. The architecture should ensure that failure of one component does not stop the entire system.

9. How would you design a secure data ingestion pipeline for a large organization?

Answer: I would design it with **encrypted data transfer, authentication, authorization, validation, and monitoring**. Data should be encrypted both in transit and at rest. Access should follow the **least-privilege principle**, while logging and auditing should track data movement. The pipeline should also include validation and error-handling mechanisms to prevent corrupted or unauthorized data from entering the system.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer: I would explain it as a system that **stores and processes huge amounts of data by distributing the workload across many computers**. Instead of depending on one powerful machine, Hadoop adds more machines when the data grows. If one machine fails, replicated data allows the system to continue working. This makes the system **scalable, reliable, and cost-effective**.

